The Remainder Extraction Program: Phase II Results

Statistical Control, Koide Blockage, and Domain Extension

Registry: [HOWL-DISC-8-2026]

Series: Discovery

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I. ABSTRACT

DISC-8 specified nine work items in three phases: statistical control, derivation, and extension. This paper reports the execution and results of all items.

Phase 1 (control) delivered two decisive results. The formal control test — 13,000 random numbers through the identical DISC-7 modular scan protocol — established that SM parameters produce hits at the same rate as random numbers (47 SM hits versus 42.3 random mean, 0/13 targets significant at $p < 0.05$). The $\alpha\_s = \pi\ \zeta(3)/32$ candidate is killed: 3.72% of random numbers near 0.118 produce the same modular signature. The VP single-threshold closure proved that no smooth change of variables can make the VP running periodic, establishing the irreducibility of the three-subgroup structure as a mathematical theorem.

Phase 2 (derivation) attempted both high-priority targets. The α_s derivation was cancelled after the control test killed its target — the candidate is noise, not physics. The Koide $a = \sqrt{2}$ derivation via frustrated graph topology produced a documented blockage: the Kuramoto model controls phase spacing while Koide requires fixed 120° spacing with variable amplitude. These are different degrees of freedom. A follow-up investigation found the midpoint reformulation: $a^2 = 2$ is the arithmetic midpoint of the positivity-allowed range [0, 4], equivalently $CV(\sqrt{m}) = 1$ is the midpoint of $[0, \sqrt{2}]$, equivalently Koide = 2/3 is the midpoint of [1/3, 1]. All three are exact Fraction identities. Quarks do not satisfy the midpoint condition (up-type CV = 1.24, down-type CV = 1.09), constraining any future derivation to be lepton-specific.

Phase 3 (extension) extracted three new physics domains — Aharonov-Bohm, flux quantization in superconductors, and the AC Josephson effect — into the remainder framework.

All three confirm Subgroup A (phase-periodic, modulus $8R_2$). The extraction table now covers 9 domains across 3 subgroups. $R_2 = \pi/4$ is present in 100% of domains. The α_s residual PSLQ scan (5 tests at maxcoeff up to 10,000) returned all null, closing the candidate from both statistical and algebraic directions.

The parameter count is unchanged: 19 → 18 confirmed (θ_QCD = 0), 18 → 17 conditional (m_τ via Koide). No new parameter was reduced. The central lesson: derivation produced all three parameter reductions in the series; scanning produced none.

II. PHASE 1: STATISTICAL CONTROL

2.1 Item 1: The Formal Control Test

PHYS-10 identified a needed test: are the DISC-7 modular scan hits distinguishable from random numbers at the same magnitudes? DISC-8 specified this as Item 1 and gated all derivation attempts on its outcome.

Protocol. For each of the 13 SM targets from DISC-7 Phase 4B, 1000 random numbers uniformly distributed in [0.9X, 1.1X] were generated (random seed 42 for reproducibility) and run through the identical modular scan: 18 moduli, remainders tested against p/q with $q \leq 20$ at 0.05% threshold. Total: 13,000 random numbers.

Results.

Target	SM Hits	Random Mean	Random Std	z-score	Significant?
α^{-1}	1	3.70	4.32	−0.62	no
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TOTAL	**47**	**42.3**	—	—	**0/13**

SM total: 47 hits. Random mean total: 42.3 hits. Zero targets exceed random expectation at $p < 0.05$.

The α_s kill. 10,000 random numbers near 0.118 were tested specifically for the $R_2 \cdot \zeta(3)$ remainder ≈ 1/8 pattern. Result: 372 hits (3.72% rate). The reason is arithmetic: $R_2 \cdot \zeta(3)$

≈ 0.942, and any number near 0.118 divided by 0.942 gives approximately $0.125 \approx 1/8$. The match is a property of the magnitude ratio, not of α_s.

Verdict: Outcome (a). The DISC-7 modular search hits are consistent with noise. The modular scan protocol at 0.05% threshold and 18 moduli has zero discriminating power between SM parameters and random numbers. The α_s = π ζ(3)/32 candidate is downgraded from "interesting but untestable" to "noise."

Script: `disc8_item1_control.py`. All assertions pass.

2.2 Item 2: VP Single-Threshold Closure

DISC-7 Phase 2 Q5 showed the full VP running cannot map to a BZ band structure because the thresholds are at unequal intervals. The question remained: could a single threshold segment, in the right coordinates, have periodic structure?

Theorem. The VP running between adjacent thresholds has no periodic structure under any smooth change of variables.

Proof. Between thresholds, $\alpha^{-1}(\kappa) = \alpha^{-1}(0) + c\kappa$ where κ = ln(μ /m_f). This is linear: $f(\kappa) = a + c\kappa$. Suppose a smooth bijection g: $\mathbb{R} \to \mathbb{R}$ makes f(g(x)) periodic with period P. Then f(g(x+P)) = f(g(x)) gives c·g(x+P) = c·g(x), hence g(x+P) = g(x) for all x. But g is bijective (injective), and a periodic function satisfies g(0) = g(P), contradicting injectivity. ∎

Corollary. Any monotonic function composed with any bijection remains non-periodic. The separation between Subgroup A (periodic) and Subgroup B (monotonic) is preserved under all smooth coordinate changes. The three-subgroup structure is irreducible — not an artifact of coordinate choice but a topological property.

The Q5 null from DISC-7 is now closed at three levels: empirical (unequal threshold intervals), structural (cosine vs logarithmic functional form), and mathematical (monotonic ∘ bijection ≠ periodic).

Script: `disc8_item2_vp_closure.py`.

III. PHASE 2: DERIVATION ATTEMPTS

3.1 Item 3: α_s Derivation (Cancelled)

The control test (Item 1) killed the α_s = π ζ(3)/32 candidate before the derivation was attempted. Random numbers match at the same rate. There is no point deriving a formula that any number near 0.118 reproduces.

The cancellation is for cause, not abandonment. The control test IS the resolution of Item 3: the target was investigated and found to be noise. No derivation is warranted for a noise artifact.

3.2 Item 4: Koide $a = \sqrt{2}$ from Frustrated Graph Topology

3.2.1 The Attempt.

The Koide formula predicts m_τ from m_e and m_mu: m_τ(Koide) = 1776.97 MeV versus measured 1776.86 ± 0.12 MeV (0.91σ). The prediction uses the parameterization $m\_i = M(1 + a\cdot\cos(\theta_0 + 2\pi i/3))^2$ with $a = \sqrt{2}$. The goal was to derive WHY $a = \sqrt{2}$.

The algebraic equivalence was proven first, as an exact Fraction identity: $a^2 = 2 \Leftrightarrow$ Koide = 2/3. The chain: the roots-of-unity identities $\Sigma\cos(\varphi\_i) = 0$ and $\Sigma\cos^2(\varphi\_i) = 3/2$ (both exact for all θ_0 at 120° spacing) give $\mathrm{sum}(m\_i)/M = 3(1 + a^2/2)$ and $(\mathrm{sum}(\sqrt{m\_i}))^2/M = 9$, producing Koide $= (1 + a^2/2)/3$. Setting this to 2/3 gives $a^2 = 2$. The converse also holds. Deriving either derives both.

3.2.2 The Blockage.

The Kuramoto model on a triangle has energy $E = -\Sigma K\_\{ij\}\cos(\varphi\_i - \varphi\_j)$. With symmetric couplings $K_{12} = K_{23} = K$ and one different bond $K_{31} = J$, the equilibrium has phase differences $\alpha = \beta$ where $\cos(\alpha) = -K/(2J)$. At K = J (all equal), $\alpha = 2\pi/3 = 120°$ (equal spacing).

The structural mismatch is fatal: the Kuramoto model controls PHASE SPACING (the departure d from 120°). The Koide parameterization has FIXED 120° spacing and varies the AMPLITUDE a. These are different degrees of freedom. Frustrating the graph moves along the wrong axis.

The generalized Koide condition with frustrated phases was computed. Setting $a^2 = 2$ and scanning the departure d from $-\pi/3$ to $\pi/3$: f(d) = 0 ONLY at d = 0. For every $d \neq 0$, the condition $a^2 = 2$ does NOT produce Koide = 2/3. The two conditions — 120° spacing and $a^2 = 2$ — are jointly required. They cannot be traded for each other.

Consequence: any future derivation attempt that tries to obtain $a = \sqrt{2}$ by modifying phase spacing will fail for the same structural reason. The path forward must explain C_3 symmetry and $a^2 = 2$ together.

Script: `disc8_item4_koide.py`. All assertions pass.

3.3 Item 4b: The CV = 1 / Midpoint Reformulation

Following the blockage, the Koide condition was reformulated in statistical language.

The CV = 1 equivalence. Define $x\_i = \sqrt{(m\_i/M)} = 1 + a\cdot\cos(\varphi\_i)$. With C_3 spacing: Mean(x) = 1, Var(x) = $a^2/2$, CV = $a/\sqrt{2}$. The chain of equivalences (all exact Fractions): Koide = 2/3 $\Leftrightarrow a^2 = 2 \Leftrightarrow$ Var(x) = 1 $\Leftrightarrow$ Var(x) = Mean(x)^2 $\Leftrightarrow$ CV = 1.

CV = 1 characterizes the exponential distribution — the maximum-entropy distribution on $[0, \infty)$ with fixed mean. However, the maximum-entropy principle for three discrete points selects all-equal masses (Koide = 1/3, the minimum), not Koide = 2/3. So maximum entropy of the mass distribution is the wrong principle.

The midpoint principle. The positivity constraint $m\_i \geq 0$ gives $a \leq 2$ (from $1 + a\cdot\cos(2\pi/3) = 1 - a/2 \geq 0$). So $a^2 \in [0, 4]$.

Three equivalent midpoint statements (all verified as exact Fraction identities):

Quantity	Range	Midpoint	Koide value
a^2	[0, 4]	2	2
CV^2	[0, 2]	1	1
Koide ratio	[1/3, 1]	2/3	2/3

The Koide ratio 2/3 is the arithmetic mean of its minimum possible value (1/3, all masses equal) and its maximum possible value (1, one mass zero).

At the extremes: $a = 0$ gives maximum symmetry (three identical masses). $a = 2$ gives maximum hierarchy (one massless generation). $a = \sqrt{2}$ sits at the exact midpoint — equally far from both extremes in a^2 space.

The quark test. Up-type quarks: CV = 1.24, Koide = 0.849. Down-type quarks: CV = 1.09, Koide = 0.731. Both above the midpoint, toward the hierarchical extreme. Only charged leptons sit at the midpoint. Any future derivation must explain why the midpoint principle is lepton-specific.

Assessment. The midpoint observation is exact, proven, and connects Koide to statistical language. It sharpens the question from "why $a = \sqrt{2}$?" to "why the midpoint of the positivity-allowed range?" But it is a mathematical restatement, not a derivation. The Koide reduction remains conditional.

Script: `disc8_item4b_cv1.py`. All assertions pass.

IV. PHASE 3: EXTENSION

4.1 Item 5: Three New Domain Extractions

Three new physics domains extracted following the DISC-7 Phase 1 protocol.

Domain 7 — Aharonov-Bohm effect. Electron encircles solenoid with flux Φ. Phase shift $\delta\varphi = 2\pi\,\Phi/\Phi_0 = 8R_2\cdot\Phi/\Phi_0$. Modulus = $2\pi = 8R_2$. At $\Phi = \Phi_0/2$, phase = $\pi = 4R_2$ (destructive interference). Verified exact. Subgroup A.

Domain 8 — Flux quantization. Superconducting ring: flux quantized from single-valuedness of the Cooper pair wavefunction. Modulus = $2\pi = 8R_2$. Remainder = 0 exactly. This is a second R = 0 mechanism within Subgroup A, distinct from θ_QCD = 0: flux quantization forces R = 0 by topology (single-valuedness), while PHYS-7 forces R = 0 by dynamics (energy minimization of cosine potential). Same result, different physics.

Domain 9 — AC Josephson effect. Voltage V produces phase accumulation $\varphi(t) = (2eV/\hbar)t$. In one Josephson period, phase accumulates exactly $2\pi = 8R_2$. The Josephson frequency-voltage relation f_J = 2eV/h is exact and is used as the international voltage standard. R_2 is embedded in the metrological definition of the volt through the same $8R_2$ modulus that appears in every other Subgroup A domain. Subgroup A.

4.2 Updated Extraction Table (9 Domains)

#	Domain	Modulus	Integer	Remainder	R_2 Role	Subgroup
1	Theta vacuum	$2\pi = 8R_2$	Instanton ν	θ = 0 (minimized)	Modulus	A
2	RG running	Mass m f	Active flavors	Running	Step $1/(12R_2)$	B
3	Bohr-Sommerfeld	$2\pi\hbar = 8R_2\hbar$	n	$\mu/4$	Modulus	A
4	Berry phase	$2\pi = 8R_2$	Winding n	γ mod 2 π	Modulus	A
5	Brillouin zone	$G = 8R_2/a$	Zone index	k mod G	Modulus	A
6	Chern-Simons	1	Chern number	CS mod $\mathbb{Z}$	Exponential	C
7	Aharonov-Bohm	$2\pi = 8R_2$	Fringe count	Phase mod 2 π	Modulus	A
8	Flux quantization	$2\pi = 8R_2$	Flux quanta n	0 (topological)	Modulus	A
9	AC Josephson	$2\pi = 8R_2$	Cycle count	Phase mod 2 π	Modulus	A

Subgroup census: A = 7 domains (theta, BS, Berry, BZ, AB, flux, Josephson). B = 1 domain (RG running). C = 1 domain (Chern-Simons). R_2 present in 9/9 (100%). Three-subgroup structure confirmed. No fourth subgroup needed.

Script: `disc8_item5_domains.py`. All assertions pass.

4.3 Item 8: α_s Residual PSLQ

The residual α_s − π ζ(3)/32 = −0.0000117 was scanned by PSLQ against the 10-constant transcendental basis. α_s itself was also scanned directly.

Test	maxcoeff	Result
Residual	100	Null
Residual	1,000	Null
Residual	10,000	Null
α s direct	1,000	Null
α s direct	10,000	Null

Five independent PSLQ tests, all null. The residual has no structure in the transcendental basis. Combined with the control test, the α_s candidate is closed from two independent directions: statistically (random numbers match at the same rate) and algebraically (no transcendental structure in the residual).

Script: `disc8_item8_residual.py`.

4.4 Items 6-7: Deprioritized

Items 6 (triple-product moduli) and 7 (scale-dependent moduli) were deprioritized after the control test showed the modular scan protocol has zero discriminating power at the tested threshold. Running more moduli through the same protocol would produce more noise. These items can be revisited only with structurally derived moduli from a physical principle, not from scanning.

4.5 Item 9: Measurement Retests (Ongoing)

The retest protocol is established. When α_s, δ_{CP}, or quark mass ratios are updated to significantly improved precision, the DISC-7 Phase 4B scan is re-run with updated values. The infrastructure is ready — only input values change. However, the control test raises the bar: even a 0.01% match (like the killed α_s candidate) is expected from random numbers. Future retests need matches at 0.001% or below, requiring 10 $\times$ or greater experimental precision improvements.

V. FALSIFICATION ASSESSMENT

Criterion	Description	Met?
F1	Control test performed	**YES —** 13,000 random numbers, full protocol
F2	α s derivation attempted	**YES —** cancelled for cause after control test killed target
F3	Koide derivation attempted	**YES —** blockage documented, midpoint reformulation produced
F4	≥6 items completed/resolved	**YES —** 5 done + 1 cancelled + 2 deprioritized = 8 resolved
F5	Both derivation items attempted	**YES —** Item 3 via control test, Item 4 directly
F6	Plan executed	**YES**

Score: 6/6. All criteria met.

VI. THE COMPLETE SEARCH NULL

With the control test, every search strategy in the series is now statistically controlled:

Strategy	Tests	Result	Control
Linear PSLQ (PHYS-10)	~600	All null	PSLQ internal calibration
Nonlinear PSLQ (DISC-7 4A)	80	All null	PSLQ internal calibration
Modular search (DISC-7 4B)	~3,960	47 hits	**Now controlled: 42.3 random = noise**
α s candidate	Specific	0.01% match	**Killed: 3.72% of randoms match**
α s residual PSLQ (DISC-8)	5	All null	PSLQ internal calibration

SM parameters do not connect to the transcendental basis through any tested relationship type at any tested precision. This is a definitive negative result within the search space covered (linear combinations with coefficients ≤ 10,000; ten nonlinear transforms with coefficients ≤ 1,000; 18 moduli with denominators ≤ 20). The search space is not exhausted but the tested region is empty.

VII. WHAT DISC-8 ACHIEVED

7.1 Statistical control. The methodological gap flagged since PHYS-10 is closed. The modular search has zero discriminating power at the tested threshold. All modular hits from DISC-7 are retroactively confirmed as noise.

7.2 Irreducibility of the three-subgroup structure. The VP closure proof establishes that Subgroup A (periodic) and Subgroup B (monotonic) cannot be unified under any smooth reparametrization. This is a mathematical theorem. The three-subgroup structure is the minimal classification, provably.

7.3 Extension to 9 domains. $R_2 = \pi/4$ is present in 100% of the 9 extracted domains. The three new domains confirm Subgroup A. No fourth subgroup is needed. Flux quantization provides a second R = 0 mechanism (topological, distinct from θ_QCD's dynamical mechanism). The Josephson effect ties R_2 to the international voltage standard.

7.4 The Koide question, sharpened. The frustrated graph mechanism is ruled out (structural mismatch: it controls phase spacing, not amplitude). The midpoint reformulation provides three equivalent exact statements of the Koide condition. The quark test shows the midpoint property is lepton-specific. The question is now: what principle selects the midpoint of the positivity-allowed range for charged lepton masses?

7.5 The central lesson. Derivation produced all three parameter reductions in the series (θ_QCD from topology, m_τ from Koide, α from QED series). Scanning produced none. The control test confirms scanning at these thresholds produces noise. The path to new reductions is derivation from physical principles, not search through transcendental combinations.

VIII. WHAT REMAINS OPEN

8.1 The Koide derivation. The conditional m_τ reduction needs a physical principle that selects $a^2 = 2$ (the midpoint). The frustrated graph path is closed. Remaining paths: a variational principle balancing symmetry against hierarchy, a generation-symmetry argument explaining why leptons (but not quarks) satisfy the midpoint condition, or a deeper understanding of the C_3 structure of three generations.

8.2 Future experimental tests. The infrastructure is ready for retesting when precision improves. The bar is high after the control test: matches must be at 0.001% or better to be distinguishable from noise.

8.3 Additional domains. Bjorken scaling (potential Subgroup B member) and the Witten index (potential Subgroup C member) were identified in the DISC-8 plan but not extracted. These remain for future work.

IX. SCRIPTS

Script	Item	Result	Assertions
disc8 item1 control.py	Control test	SM = noise (47 vs 42.3)	All pass
disc8 item2 vp closure.py	VP closure	Monotonic ≠ periodic (proof)	—
disc8 item4 koide.py	Koide derivation	Blockage: frustrated graph fails	All pass
disc8 item4b cv1.py	CV/midpoint	Koide = midpoint of [1/3, 1]	All pass
disc8 item5 domains.py	New domains	3 domains, all Subgroup A	All pass
disc8 item8 residual.py	α s residual	5/5 PSLQ null	All pass

X. THE COMPLETE SERIES

Paper	Key Result
MATH-1	$\beta = R_2 = \pi/4$ separates in 9 domains (2D)
MATH-2	17 transcendentals as integer pairs at 100 digits
MATH-3	Extended basis: elliptic integrals, Borwein $\zeta(5)$
MATH-4	Universal 2^{335} basis, 22 constants
MATH-5	R n across dimensions, n = 2, 4 uniquely binary-native
PHYS-1–4	Mass, couplings, G, test program
PHYS-5	α running at 0.02 ppm in exact arithmetic
PHYS-6	Confinement two-face
PHYS-7	θ QCD = 0 from $\mathbb{Z}$ -topology (19 → 18 confirmed)
PHYS-8	m τ from Koide, 0.91σ (18 → 17 conditional)
PHYS-9	α from a e, QED series, 4.3 ppb
PHYS-10	Remainder as observable, PSLQ null
DISC-6	Four-phase plan
DISC-7	Phases 1-3 delivered, Phase 4 null, α s candidate reported
DISC-8	**Control test kills candidate. Koide blockage documented. 9 domains. Three subgroups irreducible.**

END HOWL-DISC-8-2026

Registry: [HOWL-DISC-8-2026]

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Key Results: (1) Control test: SM modular hits = noise (47 vs 42.3 random, 0/13 significant). α_s candidate killed. (2) VP closure: monotonic ≠ periodic under any reparametrization, three subgroups proven irreducible. (3) Koide frustrated graph mechanism: documented blockage, structural mismatch. Midpoint reformulation: Koide = 2/3 = midpoint of [1/3, 1], exact. Quarks don't satisfy it. (4) Domain extension: 9 domains, R_2 100% universal, three-subgroup structure confirmed. (5) α_s residual PSLQ: 5/5 null, candidate algebraically closed.

Falsification Score: 6/6 criteria met.

Parameter Count: 18 confirmed, 17 conditional. Unchanged from DISC-7.

Central Lesson: Derivation beats search. Run the control test first.

Series: Discovery

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3.1 Item 3: α_s Derivation (Cancelled)

The control test (Item 1) killed the $\alpha_s = \pi\,\zeta(3)/32$ candidate before the derivation was attempted. Random numbers match at the same rate. There is no point deriving a formula that any number near 0.118 reproduces.

The cancellation is for cause, not abandonment. The control test IS the resolution of Item 3: the target was investigated and found to be noise. No derivation is warranted for a noise artifact.

3.2 Item 4: Koide $a = \sqrt{2}$ from Frustrated Graph Topology

3.2.1 The Attempt.

The Koide formula predicts m_τ from m_e and m_mu: m_τ(Koide) = 1776.97 MeV versus measured 1776.86 ± 0.12 MeV (0.91σ). The prediction uses the parameterization $m_i = M(1 + a\cdot\cos(\theta_0 + 2\pi i/3))^2$ with $a = \sqrt{2}$. The goal was to derive WHY $a = \sqrt{2}$.

The algebraic equivalence was proven first, as an exact Fraction identity: $a^2 = 2 \Leftrightarrow$ Koide = 2/3. The chain: the roots-of-unity identities $\Sigma\cos(\varphi_i) = 0$ and $\Sigma\cos^2(\varphi_i) = 3/2$ (both exact for all θ_0 at 120° spacing) give sum(m_i)/M = $3(1 + a^2/2)$ and $(\mathrm{sum}(\sqrt{m\_i}))^2/M = 9$, producing Koide = $(1 + a^2/2)/3$. Setting this to 2/3 gives $a^2 = 2$. The converse also holds. Deriving either derives both.

3.2.2 The Blockage.

The Kuramoto model on a triangle has energy $E = -\Sigma K_{\{ij\}}\cos(\varphi_i - \varphi_j)$. With symmetric couplings $K_{12} = K_{23} = K$ and one different bond $K_{31} = J$, the equilibrium has phase differences $\alpha = \beta$ where $\cos(\alpha) = -K/(2J)$. At K = J (all equal), $\alpha = 2\pi/3 = 120°$ (equal spacing).

The structural mismatch is fatal: the Kuramoto model controls PHASE SPACING (the departure d from 120°). The Koide parameterization has FIXED 120° spacing and varies the AMPLITUDE a. These are different degrees of freedom. Frustrating the graph moves along the wrong axis.

The generalized Koide condition with frustrated phases was computed. Setting $a^2 = 2$ and scanning the departure d from $-\pi/3$ to $\pi/3$: f(d) = 0 ONLY at d = 0. For every $d \neq 0$, the condition $a^2 = 2$ does NOT produce Koide = 2/3. The two conditions — 120° spacing and $a^2 = 2$ — are jointly required. They cannot be traded for each other.

Consequence: any future derivation attempt that tries to obtain $a = \sqrt{2}$ by modifying phase spacing will fail for the same structural reason. The path forward must explain C_3 symmetry and $a^2 = 2$ together.

Script: `disc8_item4_koide.py`. All assertions pass.

3.3 Item 4b: The CV = 1 / Midpoint Reformulation

Following the blockage, the Koide condition was reformulated in statistical language.

The CV = 1 equivalence. Define $x_i = \sqrt{(m_i/M)} = 1 + a\cdot\cos(\varphi_i)$. With C_3 spacing: Mean(x) = 1, Var(x) = $a^2/2$, CV = $a/\sqrt{2}$. The chain of equivalences (all exact Fractions): Koide = 2/3 $\Leftrightarrow a^2 = 2 \Leftrightarrow$ Var(x) = 1 $\Leftrightarrow$ Var(x) = Mean(x)^2 $\Leftrightarrow$ CV = 1.

CV = 1 characterizes the exponential distribution — the maximum-entropy distribution on $[0, \infty)$ with fixed mean. However, the maximum-entropy principle for three discrete points selects all-equal masses (Koide = 1/3, the minimum), not Koide = 2/3. So maximum entropy of the mass distribution is the wrong principle.

The midpoint principle. The positivity constraint $m_i \geq 0$ gives $a \leq 2$ (from $1 + a\cdot\cos(2\pi/3) = 1 - a/2 \geq 0$). So $a^2 \in [0, 4]$.

Three equivalent midpoint statements (all verified as exact Fraction identities):

Quantity	Range	Midpoint	Koide value
a^2	[0, 4]	2	2
CV^2	[0, 2]	1	1
Koide ratio	[1/3, 1]	2/3	2/3

The Koide ratio 2/3 is the arithmetic mean of its minimum possible value (1/3, all masses equal) and its maximum possible value (1, one mass zero).

At the extremes: $a = 0$ gives maximum symmetry (three identical masses). $a = 2$ gives maximum hierarchy (one massless generation). $a = \sqrt{2}$ sits at the exact midpoint — equally far from both extremes in a^2 space.

The quark test. Up-type quarks: CV = 1.24, Koide = 0.849. Down-type quarks: CV = 1.09, Koide = 0.731. Both above the midpoint, toward the hierarchical extreme. Only charged leptons sit at the midpoint. Any future derivation must explain why the midpoint principle is lepton-specific.

Assessment. The midpoint observation is exact, proven, and connects Koide to statistical language. It sharpens the question from "why $a = \sqrt{2}$?" to "why the midpoint of the positivity-allowed range?" But it is a mathematical restatement, not a derivation. The Koide reduction remains conditional.

Script: `disc8_item4b_cv1.py`. All assertions pass.

IV. PHASE 3: EXTENSION

4.1 Item 5: Three New Domain Extractions

Three new physics domains extracted following the DISC-7 Phase 1 protocol.

Domain 7 — Aharonov-Bohm effect. Electron encircles solenoid with flux Φ. Phase shift $\delta\varphi = 2\pi\,\Phi/\Phi_0 = 8R_2\cdot\Phi/\Phi_0$. Modulus = $2\pi = 8R_2$. At $\Phi = \Phi_0/2$, phase = $\pi = 4R_2$ (destructive interference). Verified exact. Subgroup A.

Domain 8 — Flux quantization. Superconducting ring: flux quantized from single-valuedness of the Cooper pair wavefunction. Modulus = $2\pi = 8R_2$. Remainder = 0 exactly. This is a second R = 0 mechanism within Subgroup A, distinct from θ_QCD = 0: flux quantization forces R = 0 by topology (single-valuedness), while PHYS-7 forces R = 0 by dynamics (energy minimization of cosine potential). Same result, different physics.

Domain 9 — AC Josephson effect. Voltage V produces phase accumulation $\varphi(t) = (2eV/\hbar)t$. In one Josephson period, phase accumulates exactly $2\pi = 8R_2$. The Josephson frequency-voltage relation $f_J = 2eV/h$ is exact and is used as the international voltage standard. R_2 is embedded in the metrological definition of the volt through the same $8R_2$ modulus that appears in every other Subgroup A domain. Subgroup A.

4.2 Updated Extraction Table (9 Domains)

#	Domain	Modulus	Integer	Remainder	R_2 Role	Subgroup
1	Theta vacuum	$2\pi = 8R_2$	Instanton ν	θ = 0 (minimized)	Modulus	A
2	RG running	Mass m f	Active flavors	Running	Step $1/(12R_2)$	B
3	Bohr-Sommerfeld	$2\pi\hbar = 8R_2\hbar$	n	$\mu/4$	Modulus	A
4	Berry phase	$2\pi = 8R_2$	Winding n	γ mod 2 π	Modulus	A
5	Brillouin zone	$G = 8R_2/a$	Zone index	k mod G	Modulus	A
6	Chern-Simons	1	Chern number	CS mod $\mathbb{Z}$	Exponential	C
7	Aharonov-Bohm	$2\pi = 8R_2$	Fringe count	Phase mod 2 π	Modulus	A
8	Flux quantization	$2\pi = 8R_2$	Flux quanta n	0 (topological)	Modulus	A
9	AC Josephson	$2\pi = 8R_2$	Cycle count	Phase mod 2 π	Modulus	A

Subgroup census: A = 7 domains (theta, BS, Berry, BZ, AB, flux, Josephson). B = 1 domain (RG running). C = 1 domain (Chern-Simons). R_2 present in 9/9 (100%). Three-subgroup structure confirmed. No fourth subgroup needed.

Script: `disc8_item5_domains.py`. All assertions pass.

4.3 Item 8: α_s Residual PSLQ

The residual α_s − π ζ(3)/32 = −0.0000117 was scanned by PSLQ against the 10-constant transcendental basis. α_s itself was also scanned directly.

Test	maxcoeff	Result
Residual	100	Null
Residual	1,000	Null
Residual	10,000	Null
α s direct	1,000	Null
α s direct	10,000	Null

Five independent PSLQ tests, all null. The residual has no structure in the transcendental basis. Combined with the control test, the α_s candidate is closed from two independent directions: statistically (random numbers match at the same rate) and algebraically (no transcendental structure in the residual).

Script: `disc8_item8_residual.py`.

4.4 Items 6-7: Deprioritized

Items 6 (triple-product moduli) and 7 (scale-dependent moduli) were deprioritized after the control test showed the modular scan protocol has zero discriminating power at the tested threshold. Running more moduli through the same protocol would produce more noise. These items can be revisited only with structurally derived moduli from a physical principle, not from scanning.

4.5 Item 9: Measurement Retests (Ongoing)

The retest protocol is established. When α_s, δ_CP, or quark mass ratios are updated to significantly improved precision, the DISC-7 Phase 4B scan is re-run with updated values. The infrastructure is ready — only input values change. However, the control test raises the bar: even a 0.01% match (like the killed α_s candidate) is expected from random numbers. Future retests need matches at 0.001% or below, requiring 10 $\times$ or greater experimental precision improvements.

V. FALSIFICATION ASSESSMENT

Criterion	Description	Met?
F1	Control test performed	**YES** — 13,000 random numbers, full protocol
F2	α s derivation attempted	**YES** — cancelled for cause after control test killed target
F3	Koide derivation attempted	**YES** — blockage documented, midpoint reformulation produced
F4	≥6 items completed/resolved	**YES** — 5 done + 1 cancelled + 2 deprioritized = 8 resolved
F5	Both derivation items attempted	**YES** — Item 3 via control test, Item 4 directly
F6	Plan executed	**YES**

Score: 6/6. All criteria met.

VI. THE COMPLETE SEARCH NULL

With the control test, every search strategy in the series is now statistically controlled:

Strategy	Tests	Result	Control
Linear PSLQ (PHYS-10)	~600	All null	PSLQ internal calibration
Nonlinear PSLQ (DISC-7 4A)	80	All null	PSLQ internal calibration
Modular search (DISC-7 4B)	~3,960	47 hits	**Now controlled: 42.3 random = noise**
α s candidate	Specific	0.01% match	**Killed: 3.72% of randoms match**
α s residual PSLQ (DISC-8)	5	All null	PSLQ internal calibration

SM parameters do not connect to the transcendental basis through any tested relationship type at any tested precision. This is a definitive negative result within the search space covered (linear combinations with coefficients ≤ 10,000; ten nonlinear transforms with coefficients ≤ 1,000; 18 moduli with denominators ≤ 20). The search space is not exhausted but the tested region is empty.

VII. WHAT DISC-8 ACHIEVED

7.1 Statistical control. The methodological gap flagged since PHYS-10 is closed. The modular search has zero discriminating power at the tested threshold. All modular hits from DISC-7 are retroactively confirmed as noise.

7.2 Irreducibility of the three-subgroup structure. The VP closure proof establishes that Subgroup A (periodic) and Subgroup B (monotonic) cannot be unified under any smooth reparametrization. This is a mathematical theorem. The three-subgroup structure is the minimal classification, provably.

7.3 Extension to 9 domains. $R_2 = \pi/4$ is present in 100% of the 9 extracted domains. The three new domains confirm Subgroup A. No fourth subgroup is needed. Flux quantization provides a second R = 0 mechanism (topological, distinct from θ_QCD's dynamical mechanism). The Josephson effect ties R_2 to the international voltage standard.

7.4 The Koide question, sharpened. The frustrated graph mechanism is ruled out (structural mismatch: it controls phase spacing, not amplitude). The midpoint reformulation provides three equivalent exact statements of the Koide condition. The quark test shows the midpoint property is lepton-specific. The question is now: what principle selects the midpoint of the positivity-allowed range for charged lepton masses?

7.5 The central lesson. Derivation produced all three parameter reductions in the series (θ_QCD from topology, m_τ from Koide, α from QED series). Scanning produced none. The control test confirms scanning at these thresholds produces noise. The path to new reductions is derivation from physical principles, not search through transcendental combinations.

VIII. WHAT REMAINS OPEN

8.1 The Koide derivation. The conditional m_τ reduction needs a physical principle that selects $a^2 = 2$ (the midpoint). The frustrated graph path is closed. Remaining paths: a variational principle balancing symmetry against hierarchy, a generation-symmetry argument explaining why leptons (but not quarks) satisfy the midpoint condition, or a deeper understanding of the C_3 structure of three generations.

8.2 Future experimental tests. The infrastructure is ready for retesting when precision improves. The bar is high after the control test: matches must be at 0.001% or better to be distinguishable from noise.

8.3 Additional domains. Bjorken scaling (potential Subgroup B member) and the Witten index (potential Subgroup C member) were identified in the DISC-8 plan but not extracted. These remain for future work.

IX. SCRIPTS

Script	Item	Result	Assertions
disc8 item1 control.py	Control test	SM = noise (47 vs 42.3)	All pass
disc8 item2 vp closure.py	VP closure	Monotonic ≠ periodic (proof)	—
disc8 item4 koide.py	Koide derivation	Blockage: frustrated graph fails	All pass
disc8 item4b cv1.py	CV/midpoint	Koide = midpoint of [1/3, 1]	All pass
disc8 item5 domains.py	New domains	3 domains, all Subgroup A	All pass
disc8 item8 residual.py	α s residual	5/5 PSLQ null	All pass

X. THE COMPLETE SERIES

Paper	Key Result
MATH-1	$\beta = R_2 = \pi/4$ separates in 9 domains (2D)
MATH-2	17 transcendentals as integer pairs at 100 digits
MATH-3	Extended basis: elliptic integrals, Borwein $\zeta(5)$
MATH-4	Universal 2^{335} basis, 22 constants
MATH-5	R_n across dimensions, n = 2, 4 uniquely binary-native
PHYS-1–4	Mass, couplings, G, test program
PHYS-5	α running at 0.02 ppm in exact arithmetic
PHYS-6	Confinement two-face
PHYS-7	$\theta_{QCD} = 0$ from $\mathbb{Z}$-topology (19 → 18 confirmed)
PHYS-8	m_τ from Koide, 0.91σ (18 → 17 conditional)
PHYS-9	α from a_e, QED series, 4.3 ppb
PHYS-10	Remainder as observable, PSLQ null
DISC-6	Four-phase plan
DISC-7	Phases 1-3 delivered, Phase 4 null, α_s candidate reported
DISC-8	**Control test kills candidate. Koide blockage documented. 9 domains. Three subgroups irreducible.**

END HOWL-DISC-8-2026-FINAL

Registry: [?]

Status: Complete

Domain: Research Program / Mathematical Physics

Key Results: (1) Control test: SM modular hits = noise (47 vs 42.3 random, 0/13 significant). α_s candidate killed. (2) VP closure: monotonic ≠ periodic under any reparametrization, three subgroups proven irreducible. (3) Koide frustrated graph mechanism: documented blockage, structural mismatch. Midpoint reformulation: Koide = 2/3 = midpoint of [1/3, 1], exact. Quarks don't satisfy it. (4) Domain extension: 9 domains, R_2 100% universal, three-subgroup structure confirmed. (5) α_s residual PSLQ: 5/5 null, candidate algebraically closed.

Falsification Score: 6/6 criteria met.

Parameter Count: 18 confirmed, 17 conditional. Unchanged from DISC-7.

Central Lesson: Derivation beats search. Run the control test first.

[**HOWL-DISC-8-2026**] The Remainder Extraction Program: Phase II Results. Github: https://github.com/ghowland/papers/tree/main/papers/DISC/HOWL-DISC-8-2026

[**HOWL-DISC-6-2026**] The Remainder Extraction Program. Github: https://github.com/ghowland/papers/tree/main/paper DISC-6-2026

[**HOWL-DISC-7-2026**] The Remainder Extraction Program: Execution and Results. Github: https://github.com/ghowland/papers/tree/main/papers/DISC/HOWL-DISC-7-2026